Hierarchical Convexity-Run Decomposition: Finite-Sample Chord-Lobe Recovery and Interpretable LC–MS Transfer

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Abstract

We introduce Hierarchical Convexity-Run Decomposition (HCRD), a finite nonlinear transform that replaces runs of one discrete-curvature sign by endpoint chords and recurses on retained knots. Signed residuals form an exactly reconstructing, interpretable lobe hierarchy with linear total knot-only work. For a non-circular signal class with alternating curvature blocks and isolated zero-curvature joins, we prove finite-sample boundary and component recovery under Gaussian noise. If the minimum active curvature γ exceeds twice the simultaneous threshold τ, every first-level boundary is recovered with probability at least $1-\delta$, with explicit baseline and detail error bounds. A 44,000-draw phase diagram across four event-count/width configurations followed the predicted transition: all eight cells with $\gamma/\tau > 2$ had exact-recovery rate 1.000 (smallest 95% Wilson lower bound 0.996). Structural results further establish finite termination, affine equivariance, variation control, the hard map's global discontinuity, and stable companions. In frozen no-refit transfers between two fully labelled LC–MS studies, HCRD-8 plus an independently reimplemented quality score improved average precision by 0.1181 and 0.1005 (Holm-adjusted $p = 0.00040$ in both). An equal-dimensional Gaussian-derivative control tied HCRD in one direction, whereas HCRD was superior in the reverse; a larger TARDIS/FAME shift did not confirm superiority. HCRD therefore supplies a simple morphological inductive bias with a stated recovery regime and explicit application boundaries.

1 Introduction

Empirical mode decomposition (EMD) motivated data-adaptive analysis of nonstationary signals through extrema and envelope sifting [19]. Intrinsic time-scale decomposition (ITD) subsequently introduced an efficient iterative baseline–rotation construction based on piecewise-linear transformations between extrema [16, 17]. In particular, its patent measures a lobe height from an extremum to the line joining neighbouring extrema and recursively decomposes the concatenated baseline. Thus straight-line lobe geometry and recursive piecewise-linear baselines are prior art. The method studied here arose independently from a different selection and update rule: partition a sampled graph into maximal runs of one divided-curvature sign, replace each entire run by its endpoint chord, and allow only the retained knots at the next level.

The primitive is related to inflection-point sampling [20], but iteration produces a nested, exactly reconstructing hierarchy. More directly, the PDE sifting construction of Deléchelle et al. [10] converges to a piecewise-cubic interpolant through successive inflection points, while the α-shape construction of Chessel et al. [5] defines a convexity-driven geometric scale space. Recent EMD software also includes inflection-point interpolation as one of several local-mean estimators [40], and special-inflection-point EMD augments extrema with inflections from the signal and its derivatives [22]. These works preclude treating the use of inflection points itself as novel. HCRD differs in using a one-pass discrete curvature-sign walk, endpoint chords, and recursively nested eligible knot sets, rather than auxiliary envelope knots, PDE evolution, α-shape subsampling, or repeated IMF sifting.

The closest theoretical comparison classes further include Gaussian scale space [3], ℓ_1 trend filtering [23], iterative filtering [29], connected operators represented by region trees [35], and the LULU discrete pulse transform and its linear-time one-dimensional Roadmaker construction [1, 27]. Connected operators prune flat-zone region trees, while DPT produces constant block pulses on connected supports; neither uses affine residual lobes from the HCRD curvature-run knot tree. HCRD's contribution is the particular finite discrete operator and the combination of hierarchy, reconstruction, geometric sign, stability, limitation, and calibrated-inference results—not the generic idea of using curvature, inflection points, chord lobes, or a nonlinear component hierarchy.

Table 1 makes the closest distinctions explicit. ITD is the nearest chord-baseline recursion, whereas DPT is the nearest exact nonlinear pulse hierarchy; neither uses the HCRD curvature-run knot rule. Trend filtering supplies the closest globally stable piecewise-linear estimator, but it solves a penalized optimization problem rather than a finite run walk.

Table 1: Conceptual comparison with the closest one-dimensional methods. "Exact" means that the native components plus the final baseline reconstruct the input. Costs are nominal algorithmic costs, not the runtime experiment.

Method	Selection primitive	Baseline/detail	Exact	Nested	Nominal work	Stability/grid
HCRD	curvature-sign runs	endpoint chords	yes	knot subsets	$O(n)$ total sparse	hard-discontinuous; irregular x
ITD [17]	extrema updates	linear baseline/rotations	yes	recursive baseline	$O(n)$ per level	hard extrema; ordered grid
LULU/DPT [1, 27]	order/flat zones	constant pulses	yes	size hierarchy	$O(n)$ Roadmaker	order-based sequence
RDP [11]	maximum chord error	polygonal approximation	with residual	split tree	$O(n^2)$ worst case	hard ties; arbitrary coordinates
ℓ_1 trend filtering [23]	penalized curvature	piecewise-linear fit	with residual	not generally	convex optimization	continuous fit; irregular variants
Gaussian scale space [3]	convolution scale	smoothed signals	scale differences	scale-ordered	FFT/direct filtering	continuous; regular-grid standard

The LC–MS use case is also distinct from end-to-end peak picking. PeakBot detects local maxima and classifies two-dimensional chromatographic regions with a convolutional network [4]; PeakDetective combines an autoencoder with active learning for dataset-specific curation [38]; TMSQE learns targeted peak-quality embeddings [42]; and MassCube integrates feature detection, evaluation, grouping, and annotation [43]. HCRD instead supplies transparent features for already bounded one-dimensional EICs and is tested here under no-target-refit transfer. These systems therefore define the application landscape but are not interchangeable baselines for the present fixed-box question.

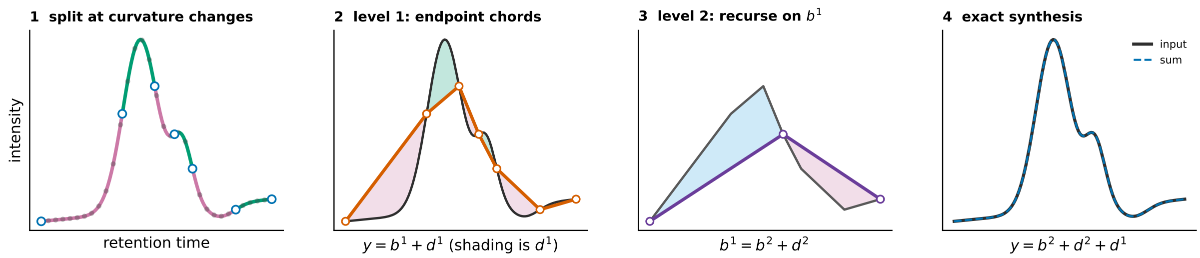

Figure 1: Two explicit HCRD steps on a synthetic EIC-like graph. Curvature-sign changes delimit retained knots. Replacing every level-1 run by its endpoint chord gives baseline b^1 and signed residual d^1; applying the same operation to b^1 gives b^2 and d^2. The displayed truncation reconstructs $y = b^2 + d^2 + d^1$ exactly, and recursion may continue. Shaded lobes can be summarised by area or energy, but the decomposition is the complete residual shape. This computed example illustrates the operator; it is not a stability proof and levels are not asserted to be frequency bands.

Our contributions are:

1. a complete finite algorithm for the centred curvature-run operator, with nested-hierarchy, exact-reconstruction, sparse linear-work, affine, variation, and interpretable lobe-shape results;

2. an HCRD-specific finite-sample recovery theorem for an explicit alternating chord-lobe class, local and stochastic hierarchy-agreement guarantees, and a theorem-linked 44,000-draw phase diagram;

3. a sharp global discontinuity boundary and stable proximal-curvature/persistence companions;

4. supporting inference for fixed or independently guided HCRD-shaped templates: finite-sample scans, a matching-order $\sqrt{\log M}$ benchmark over an orthonormal subclass, projected-norm control for a continuous family, and replicate/buffered/tree procedures under explicit independence assumptions;

5. frozen cross-study LC–MS validation, multilevel and area ablations, a matched-capacity conventional control, operational review-queue analysis, and external limitation studies.

2 Definition of the transform

Let $x_0 < \cdots < x_{n-1}$ and $y = (y_0, \ldots, y_{n-1}) \in \mathbb{R}^n$. Define divided slopes and discrete curvatures by

$$s_i(y) = \frac{y_{i+1} - y_i}{x_{i+1} - x_i}, \qquad \kappa_i(y) = s_i(y) - s_{i-1}(y). \tag{1}$$

For absolute and relative tolerances $a, r \geq 0$, define

$$\tau = a + r \max\{1, \max_i |s_i(y)|\}. \tag{2}$$

A curvature is active when $|\kappa_i| > \tau$ and inactive otherwise. The mathematical hard operator uses $a = r = 0$; the implementation defaults to a tiny relative tolerance to suppress floating-point curvature on interpolated affine segments. Algorithm 1 specifies every branch of the centred rule.

Algorithm 1 One centred HCRD hierarchy on an irregular grid

Input: strictly increasing x, samples y, tolerances $a, r \geq 0$, and optional depth cap L. Initially $E = (0, \ldots, n-1)$.

1. On the current eligible list $E = (e_0, \ldots, e_{m-1})$, set $z_i = y_{e_i}$, $u_i = x_{e_i}$, compute divided curvatures, and compute τ from the current divided slopes. If $m = 2$, stop.
2. Set the local start $p = 0$ and the new local knot list $K = (0)$.
3. Let f be the first active curvature index in $\{p+1, \ldots, m-2\}$. If none exists, append $m-1$ to K and finish this level. Otherwise set the active sign to $\operatorname{sign}(\kappa_f)$, $\ell = f$, and scan right.
4. Ignore inactive curvatures. On each active curvature of the same sign, update ℓ. Let q be the first active curvature of the opposite sign. If no such q exists, append $m-1$ and finish the level.
5. If $q - \ell = 2$, the single inactive curvature at $\ell + 1$ is the sampled transition and the candidate is $c = \ell + 1$. At zero tolerance this is exactly the sampled-zero case.
6. Otherwise set $c = \ell$ only when $|\kappa_\ell| < |\kappa_q|$ and $\ell \neq f$; set $c = q$ in every other case. Thus equal magnitudes are tied to the right, first-opposite side, and a long inactive plateau never creates an interior knot.
7. Enforce strict coarsening by replacing c with $\max\{p+2, \min(c, m-1)\}$. Append c to K, set $p = c$, and return to step 3. The final remainder always closes at $m-1$.
8. Map the local knots back to original indices, $E' = (e_k : k \in K)$. Interpolate the endpoint chords through (x_i, y_i) for $i \in E'$ to obtain the next baseline; subtract it to obtain the signed detail.
9. Set $E \leftarrow E'$ and repeat until $|E| = 2$ or the optional cap L is reached. Later levels therefore use only previously retained knots.

Let $K(y)$ denote the first mapped knot set and let Ty be linear interpolation through (x_k, y_k) for $k \in K(y)$.

Set $b^0 = y$ and recursively define

$$b^{j+1} = Tb^j, \qquad d^{j+1} = b^j - b^{j+1}. \tag{3}$$

The implementation restricts eligible knots at level $j+1$ to knots retained at level j. This prevents the insertion of geometrically redundant samples inside an existing affine segment. In local eligible-knot coordinates, a nonterminal centred interval is required to advance by at least two old intervals; the first-opposite and sampled-zero cases below satisfy this convention explicitly.

3 Structural results

Theorem 1 (Exact reconstruction)**.** *For every computed level J,*

$$y = \sum_{j=1}^{J} d^j + b^J. \tag{4}$$

Proof. Substitute $d^j = b^{j-1} - b^j$ and telescope. □

Theorem 2 (Finite nested hierarchy)**.** *The eligible knot sets are nested. Every nonterminal HCRD interval consumes at least two intervals of the preceding level. Consequently the hierarchy ends at the endpoint chord after at most* $\max\{1, \lceil \log_2(n-1) \rceil\}$ *implemented levels.*

Proof. Eligibility gives nesting. A run can close only after the walk has passed the first active old curvature and reached either an eligible intervening zero or the first active curvature of opposite sign. Thus an old interior knot is removed in each nonterminal new interval. Pairing old intervals gives the halving recurrence, with at most one terminal remainder. □

Corollary 1 (Linear-work knot-only hierarchy)**.** *Let $N_0 = n-1$ and let N_j be the number of intervals after level j. An implementation that scans only the current eligible knot list has $O(n)$ total knot-walk work and stores*

$$\sum_{j=1}^{J} |K_j| < 2(n-1) + J = O(n). \tag{5}$$

Dense length-n baselines and details can be reconstructed exactly on demand.

Proof. The halving recurrence gives $N_j \le \lceil N_0/2^j \rceil$ and $J = O(\log n)$. For integer $N_0 \ge 1$, $\lceil N_0/2^j \rceil \le (N_0 - 1)/2^j + 1$; hence $\sum_{j=1}^{J} N_j < N_0 - 1 + J \le 2N_0$. Since $|K_j| = N_j + 1$, the displayed finite bound follows. The analogous series beginning with N_0 bounds the total scanned eligible intervals by $O(n)$. Chord interpolation through each stored knot set reproduces the dense baseline, and adjacent differences reproduce the details. □

Proposition 1 (Signed structures)**.** *On a discretely convex run, $y - Ty \le 0$. On a discretely concave run, $y - Ty \ge 0$.*

Proof. A convex polygonal graph lies below its endpoint secant. Reverse the inequality for a concave graph. □

For structure r of detail d^j on $I_{jr} = [x_a, x_b]$, define its signed area, geometric mass, and quadratic residual energy by

$$S_{jr} = \int_{I_{jr}} d^j(x)\,dx, \qquad A_{jr} = \int_{I_{jr}} |d^j(x)|\,dx, \qquad E_{jr} = \int_{I_{jr}} d^j(x)^2\,dx, \tag{6}$$

where d^j is the exact piecewise-linear interpolant of the sampled residual. These quantities are computable from retained knots without materializing a length-n detail. Signed structures give $A_{jr} = |S_{jr}|$. Under $y \mapsto cy + \alpha + \beta x$, A_{jr} scales by $|c|$ and E_{jr} by c^2; under an orientation-preserving horizontal affine change $x \mapsto \rho x + \tau$, both integrals acquire the Jacobian factor ρ. Thus A_{jr} is a literal morphology measure in signal-units times coordinate-units. It is not spectral energy, and the nonorthogonal hierarchy has no Parseval identity.

Proposition 2 (Dimensionless lobe shape)**.** *For a nonzero structure let $W = x_b - x_a$, $H = \max_{I_{jr}} |d^j|$, and define*

$$F = \frac{A_{jr}}{WH}, \qquad C = \frac{A_{jr}^2}{WE_{jr}}. \tag{7}$$

Then $0 < F \le 1$ and $0 < C \le 1$. Both are invariant under nonzero vertical scaling, affine trend addition, and orientation-preserving horizontal affine changes. A piecewise-linear triangular lobe has $(F, C) = (1/2, 3/4)$ regardless of apex location; a parabolic lobe $4Ht(W-t)/W^2$ has $(F, C) = (2/3, 5/6)$.

Proof. $A \le WH$ follows from $|d| \le H$, while Cauchy–Schwarz gives $A^2 \le WE$. The scaling laws above cancel in both ratios. Direct integration gives the two examples. □

The first ratio is the implemented per-structure shape factor. The second is available as a quadratic concentration descriptor but was not added after the LC–MS protocols were frozen. Keeping A_{jr} or either invariant for each moving support and level also yields a low-dimensional temporal morphology series; collapsing all structures to one total mass discards that evolution.

Theorem 3 (Affine equivariance)**.** *For zero curvature tolerance, nonzero a, and $\ell_i = \alpha + \beta x_i$,*

$$T(ay + \ell) = aTy + \ell. \tag{8}$$

Hence every HCRD detail is invariant to affine trend addition and scales by a.

Proof. Affine addition cancels from adjacent divided-slope differences. Multiplication by a preserves all transition locations, although negative a reverses the signs. Chord interpolation commutes with both operations. □

Theorem 4 (Range and total variation)**.** *For one HCRD step,*

$$\min y \le (Ty)_i \le \max y, \qquad TV(Ty) \le TV(y). \tag{9}$$

Proof. Every chord sample is a convex combination of its endpoint values. Its variation equals the absolute difference between the endpoint values, which is no greater than the variation of the removed polygonal subpath. □

4 Stability and its obstruction

Theorem 5 (Stability inside a curvature-decision cell)**.** *On a unit grid suppose $\gamma = \min_i |\Delta^2 y_i| > 0$ and let*

$$\eta = \min_{\Delta^2 y_i \Delta^2 y_{i+1} < 0} \big| |\Delta^2 y_i| - |\Delta^2 y_{i+1}| \big|, \tag{10}$$

with $\eta = +\infty$ if there is no sign transition. For the centred rule, if $\|e\|_\infty < \min\{\gamma/4, \eta/8\}$, then $K(y+e) = K(y)$ and

$$\|T(y+e) - Ty\|_\infty \le \|e\|_\infty, \tag{11}$$

$$\|(I-T)(y+e) - (I-T)y\|_\infty \le 2\|e\|_\infty. \tag{12}$$

Proof. The coefficient norm of the second-difference stencil $(1, -2, 1)$ is four, so the first margin prevents a curvature-sign change. A difference of two absolute curvature magnitudes moves by at most $8\|e\|_\infty$, so the second margin preserves every centred transition comparison. The knot walk is therefore fixed. Fixed-knot interpolation is a convex combination of endpoint perturbations, and the detail estimate follows by the triangle inequality. □

The second margin is necessary: $(0,0,1,3,4)$ has curvatures $(1,1,-1)$, whereas $(0,0,1,3-\varepsilon, 4-2\varepsilon)$ has the same signs but selects the other side of the tied centred transition. The legacy first-opposite rule does not compare magnitudes and requires only the sign margin.

Proposition 3 (Global discontinuity)**.** *The raw HCRD baseline operator is not globally continuous.*

Proof. For $y^\pm = (-2, 0, 2 \pm \varepsilon, -2)$, the inputs approach one another as $\varepsilon \downarrow 0$, whereas one first baseline retains the interior peak and the other becomes the constant endpoint chord. Their sup-norm separation tends to four. □

Theorem 6 (No continuous generic-agreement extension)**.** *For four-sample signals let $G = \{y : \min_i |\Delta^2 y_i| > 0\}$. There is no globally continuous map $F : \mathbb{R}^4 \to \mathbb{R}^4$ that agrees with the raw HCRD baseline T on all of G.*

Proof. For every $0 < \varepsilon < 1$, both $y_\varepsilon^\pm = (-2, 0, 2 \pm \varepsilon, -2)$ lie in G and converge to the same y^0. Agreement on G makes $F(y_\varepsilon^\pm)$ equal the two HCRD outputs in the preceding counterexample, whose limits are separated by four. Continuity at y^0 would force both limits to equal $F(y^0)$, a contradiction. □

Remark 1. *HCRD is not an orthogonal transform and need not contract Euclidean energy. For $y = (-2, -1, -2)$, one has $Ty = (-2, -2, -2)$ and $\|Ty\|_2 > \|y\|_2$.*

5 A stable proximal companion

Let D be the divided-curvature matrix and let ϕ be a proper closed convex function. Define

$$P_\phi y = \operatorname*{argmin}_z \left\{ \frac{1}{2}\|y - z\|_2^2 + \phi(Dz) \right\}, \qquad Q_\phi = I - P_\phi. \tag{13}$$

We implement the established ℓ_1 trend penalty $\phi(u) = \lambda\|u\|_1$ [23] and a quadratic curvature penalty $\phi(u) = \lambda\|u\|_2^2/2$. These guide operators themselves are not claimed as novel; their role is to give HCRD an exact stable outer split.

Theorem 7 (Exact globally nonexpansive outer split)**.** *Both P_ϕ and Q_ϕ are firmly nonexpansive and*

$$y = P_\phi y + Q_\phi y. \tag{14}$$

Moreover, if $D\ell = 0$, then $P_\phi(y + \ell) = P_\phi y + \ell$ and $Q_\phi(y+\ell) = Q_\phi y$.

Proof. The divided-curvature matrix has full row rank, so properness of ϕ makes $\phi \circ D$ proper. The objective is coercive and strongly convex, hence has a unique minimizer. The standard proximal-map theorem and Moreau decomposition make a proximal map and its residual firmly nonexpansive [33]. Exact splitting is the definition of Q_ϕ. Translating both y and the candidate minimizer by $\ell \in \ker D$ leaves the objective unchanged. □

Corollary 2 (Irregular-grid sign certificate)**.** *Suppose $\|e\|_2 \le \rho$ and put $h_{i-1} = x_i - x_{i-1}$ and $h_i = x_{i+1} - x_i$. The ith guide curvature changes by at most*

$$c_i\rho = 2\rho(h_{i-1}^{-1} + h_i^{-1}). \tag{15}$$

Thus every guide curvature satisfying $|\kappa_i(P_\phi y)| > c_i\rho$ has a fixed sign throughout the input ball. Other curvatures must be marked uncertified.

Proof. Theorem 7 bounds the guide perturbation in ℓ_2, hence also coordinatewise, by ρ. The absolute coefficient sum of the irregular three-point curvature row is $2(h_{i-1}^{-1} + h_i^{-1})$. □

We retain $Q_\phi y$ explicitly and apply HCRD only to the guide $P_\phi y$, so the complete representation still reconstructs y. Theorem 7 applies to the outer split, not to the hard knot hierarchy. Indeed, Theorem 6 rules out global continuity for any hard replacement that preserves every generic raw decision.

5.1 Stable signed-curvature persistence

Hard knot locations are not the only possible summary of the curvature walk. Let P be the path on the $n-2$ interior sample indices and define the nonnegative vertex functions

$$u_y^+ = (D_x y)_+, \qquad u_y^- = (-D_x y)_+. \tag{16}$$

Extend them linearly over the path edges. For a nonnegative function u, let $\mathcal{D}_0^{\mathrm{fin}}(u)$ be the finite part of its ordinary zero-dimensional superlevel persistence diagram and let $M(u) = \max u$ record the birth of its unique essential component. Define

$$\begin{aligned} d_{\mathrm{SC}}(y,z) = \max\{&d_B(\mathcal{D}_0^{\mathrm{fin}}(u_y^+), \mathcal{D}_0^{\mathrm{fin}}(u_z^+)), |M(u_y^+) - M(u_z^+)|,\\ &d_B(\mathcal{D}_0^{\mathrm{fin}}(u_y^-), \mathcal{D}_0^{\mathrm{fin}}(u_z^-)), |M(u_y^-) - M(u_z^-)|\}. \end{aligned} \tag{17}$$

Theorem 8 (Global signed-curvature persistence stability)**.** *For two signals on the same irregular grid,*

$$d_{\mathrm{SC}}(y,z) \le C_x\|y-z\|_\infty, \qquad C_x = 2\max_i\{h_{i-1}^{-1} + h_i^{-1}\}. \tag{18}$$

If $\|y-z\|_\infty \le \varepsilon$, every finite bar of either sign with lifetime greater than $2C_x\varepsilon$ has a finite same-sign match for the other signal.

Proof. The absolute row sum of D_x is $2(h_{i-1}^{-1} + h_i^{-1})$, while positive and negative part maps are 1-Lipschitz. Hence $\|u_y^\pm - u_z^\pm\|_\infty \le C_x\|y-z\|_\infty$. The finite path is triangulable and the piecewise-linear functions are tame, so classical persistence-diagram stability applies [7]. The unique essential classes must match each other; the remaining matching is between finite points. A finite bar of lifetime L costs $L/2$ to match to the diagonal, proving the last statement. □

Peak indices are retained by the implementation as interpretive metadata but are not included in d_{SC}. Two distant curvature peaks of equal height can exchange the essential label under an arbitrarily small perturbation, so a global coordinate-stability statement would be false. Thus Theorem 8 is a stable lobe representation, not a concealed claim that the hard HCRD knots are continuous.

Proposition 4 (Curvature visibility limit)**.** *Let $y(t) = g(t) + a\sin(\omega t)$ on an interval and suppose $g''(t) \ge c > |a|\omega^2$ throughout that interval. Then $y''(t) > 0$ there, so the sinusoidal component creates no curvature-sign transition and cannot be isolated by a raw inflection-run rule on that interval.*

Proof. $y''(t) = g''(t) - a\omega^2\sin(\omega t) \ge c - |a|\omega^2 > 0$. Thus y is strictly convex; its divided slopes on every increasing sampled grid are increasing, so its discrete curvatures have one positive sign. □

6 A simultaneous noise-aware modification

Assume a uniform grid with spacing h and observations $Y_i = y_i + \varepsilon_i$, where the errors are independent $N(0, \sigma^2)$ variables.

Theorem 9 (Family-wise curvature threshold)**.** *For $0 < \delta < 1$, let*

$$\tau = \frac{\sigma}{h}\sqrt{12 \log \frac{2(n-2)}{\delta}}. \tag{19}$$

With probability at least $1-\delta$, every observed curvature differs from its population value by at most τ. On this event, population zero curvatures are thresholded to zero and all curvatures with magnitude greater than 2τ retain their correct active sign.

Proof. Each curvature error has variance $6\sigma^2/h^2$. Apply a Gaussian tail bound to each of the $n-2$ entries and take a union bound; independence of curvature errors is not required. The classification statement follows from the triangle inequality. □

For a common physical-unit alternative to the row-normalized anomaly score, let $h_{\min} = \min_i(x_{i+1} - x_i)$,

$$\epsilon = \sigma\sqrt{2\log(2n/\delta)}, \qquad \eta = 4\epsilon/h_{\min}, \qquad \tau = \lambda\eta, \quad \lambda \geq 1, \tag{20}$$

run HCRD with absolute curvature tolerance τ, and set

$$Z(t) = \left[\max_j |d_j(t)| - 2\epsilon\right]_+. \tag{21}$$

Theorem 10 (Common-scale affine-null certificate)**.** *If $Y_i = \alpha + \beta x_i + e_i$ with independent $e_i \sim N(0, \sigma^2)$, then $\Pr\{Z(t) = 0$ for every $t\} \geq 1 - \delta$.*

Proof. A union bound gives $\|e\|_\infty \leq \epsilon$ with probability at least $1-\delta$. On every retained grid a divided slope changes by at most $2\epsilon/h_{\min}$, so a curvature changes by at most $\eta \leq \tau$. The affine mean has zero curvature; the thresholded hierarchy therefore keeps only the endpoints. Both a noisy sample and its noisy endpoint chord lie within ϵ of their affine counterparts, hence every density is at most 2ϵ. □

Theorem 11 (Multilevel hierarchy-agreement margins)**.** *Run the noiseless minimum-curvature hierarchy with tolerance τ through level L. At every visited level suppose each curvature declared inactive satisfies $|\kappa_i| + \eta \leq \tau$, each active curvature satisfies $|\kappa_i| - \eta > \tau$, and each comparison $|\kappa_i| < |\kappa_j|$ used to choose a side of an unsampled sign transition has gap $\big||\kappa_i| - |\kappa_j|\big| > 2\eta$. Then, with probability at least $1-\delta$, the noisy and noiseless knot sets agree through level L. Consequently a retained noiseless chord lobe of height $H > 4\epsilon$ has strictly positive score at its noiseless peak on that event.*

Proof. On $\|e\|_\infty \leq \epsilon$, every active-grid curvature moves by at most η. The first two margins preserve threshold status and sign; the final margin preserves every magnitude comparison. Thus the complete first-level branch trace is unchanged. Induction applies because equal knot sets expose the same retained grid at the next level. Under agreement, a density changes by at most 2ϵ; subtracting the null radius gives the peak lower bound $[H - 4\epsilon]_+$. □

Theorem 12 (Stochastic hierarchy agreement)**.** *Fix a grid, tolerance τ, and depth L. For a latent signal f, replay all threshold and centred magnitude-comparison decisions visited by its noiseless hierarchy. Let $M_{\mathrm{th}}(f)$ be the smallest visited distance between a curvature magnitude* *and τ, let $M_{\mathrm{cmp}}(f)$ be the smallest visited gap between compared curvature magnitudes, and define the sufficient input decision radius*

$$R(f) = \frac{h_{\min}}{4} \min\{M_{\mathrm{th}}(f), M_{\mathrm{cmp}}(f)/2\}, \tag{22}$$

with an empty minimum interpreted as $+\infty$. If random F satisfies $\Pr\{R(F) \leq r\} \leq H(r)$ and $e \sim N(0, \sigma^2 I_n)$ is independent of F, then

$$\Pr\{K_\ell(F+e) = K_\ell(F),\ 1 \leq \ell \leq L\} \geq \left[1 - H(r) - 2n \exp\{-r^2/(2\sigma^2)\}\right]_+ . \tag{23}$$

The bound holds for every $r > 0$ and may therefore be optimized over r.

Proof. A sup-norm perturbation u changes every visited divided curvature by at most $4u/h_{\min}$ and a compared magnitude gap by at most twice that quantity. Thus $\|e\|_\infty < R(F)$ preserves the complete visited branch trace, and induction preserves every knot set. Agreement holds on $\{R(F) > r\} \cap \{\|e\|_\infty < r\}$; a union bound and the Gaussian maximum tail give the result. □

Corollary 3 (Transverse curved-background–lobe populations). *Suppose $F = F_\Theta$, where Θ has density at most $p_{\max}$ on a compact parameter box. Partition the box into finitely many hierarchy branch cells and write the signed decision margins, already scaled to input-radius units in the definition of R, as $m_j(\theta)$. If each m_j is C^1 on its cells, $\|\nabla m_j\|_2 \geq g_j > 0$ in $|m_j| \leq r_0$, and every level set $m_j^{-1}(u)$ has $(d-1)$-measure at most A_j for $|u| \leq r_0$, then*

$$H(r) = 2p_{\max} r \sum_j A_j/g_j, \qquad 0 < r \leq r_0, \tag{24}$$

is valid in Theorem 12.

Proof. The coarea formula bounds the probability of the r-tube around decision surface j by $2p_{\max} r A_j/g_j$; union-bound the finite visited margin list over the branch cells. □

The radius is sufficient, not largest. The corollary covers random lobe amplitude, location, width and asymmetry together with finitely parameterised curved backgrounds when decision surfaces are transverse. Atoms on a tie, tangencies, infinite-dimensional backgrounds, and same-data estimation of H remain outside the result.

7 Finite-sample recovery on a chord-lobe class

On a uniform grid $x_i = x_0 + ih$, define the divided-slope curvature

$$\kappa_i(z) = \frac{z_{i+1} - 2z_i + z_{i-1}}{h}, \qquad 1 \leq i \leq n-2. \tag{25}$$

This scaling matches Algorithm 1; it is a slope difference rather than a divided second derivative.

Theorem 13 (Finite-sample generative chord-lobe recovery). *Let $0 = k_0 < k_1 < \cdots < k_J = n-1$, with $k_r - k_{r-1} \geq 2$, and write $f = b + q$. Suppose b is affine on each $[x_{k_{r-1}}, x_{k_r}]$, $q_{k_r} = 0$, and the population curvatures of f satisfy*

1. *$\kappa_{k_r}(f) = 0$ at every interior knot;*
2. *inside successive knot intervals they have alternating constant sign and magnitude at least $\gamma > 0$.*

Observe $Y_i = f_i + e_i$ with independent $e_i \sim N(0, \sigma^2)$ and fix $0 < \delta < 1$. Set

$$\tau_\delta = \frac{\sigma}{h}\sqrt{12 \log \frac{4(n-2)}{\delta}}, \qquad \epsilon_\delta = \sigma\sqrt{2 \log \frac{4n}{\delta}}. \tag{26}$$

Run the first centred HCRD step with absolute curvature tolerance τ_δ. If $\gamma > 2\tau_\delta$, then with probability at least $1-\delta$ its knot set is exactly $\{k_0, \ldots, k_J\}$. On the same event, for its chord baseline $\widehat{b}$ and detail $\widehat{q} = Y - \widehat{b}$,

$$\|\widehat{b} - b\|_\infty \le \epsilon_\delta, \quad n^{-1}\|\widehat{b} - b\|_2^2 \le \epsilon_\delta^2, \quad \|\widehat{q} - q\|_\infty \le 2\epsilon_\delta, \quad n^{-1}\|\widehat{q} - q\|_2^2 \le 4\epsilon_\delta^2. \tag{27}$$

Proof. With probability at least $1 - \delta/2$, every curvature error is at most τ_δ: this is Theorem 9 with its error budget replaced by $\delta/2$. Population zeros are then inactive, whereas $\gamma > 2\tau_\delta$ makes every active curvature remain active with the correct sign. Hence each declared knot is the unique inactive eligible point between opposite active blocks. The centred transition rule selects that point, and the two-segment gap prevents the strict-coarsening correction from moving it. Induction along the first-level walk proves exact knot recovery.

Independently, a Gaussian maximum bound gives $\|e\|_\infty \le \epsilon_\delta$ with probability at least $1-\delta/2$. On exact recovery, $f_{k_r} = b_{k_r}$ and b is its own chord interpolant, so $\widehat{b} - b$ is the chord interpolant of the knot errors. Its sup norm is at most $\|e\|_\infty$; subtracting it from e gives the detail bound. The MSE bounds follow from the sup-norm bounds, and a union bound completes the proof. □

The assumptions are structural rather than circular: they describe sampled curvature before running HCRD. An explicit subclass has $k_r = rm$, a globally affine b, and equal alternating parabolic lobes

$$q_i = s(-1)^r 4A u_i(1-u_i), \qquad u_i = (i - rm)/m, quadrm \le i \le (r+1)m. \tag{28}$$

Here every join has zero sampled curvature and $\gamma = 8A/(m^2 h)$. Unequal adjacent amplitudes can destroy the zero join and move a boundary by one sample, so the theorem does not silently extend to that case. The noiseless version is exact for every $A > 0$; the noisy guarantee is a sufficient, not necessary, separation condition.

8 Template scans for chord lobes

These are conservative sup-norm certificates. They are not sharp detection boundaries when location and scale are unknown. Modern multiscale shape tests and scan statistics provide the relevant calibration and minimax comparison targets [14, 2, 36].

The complete HCRD shapes also admit a sharper finite-dictionary statement. Let P project onto $\mathrm{span}\{\mathbf{1}, x\}$ and let $M \ge 2$ and $v_1, \ldots, v_M \in \mathrm{range}(I - P)$ be unit-norm, oriented lobe templates. They may be fixed in advance or extracted from an independent guide replicate; in the latter case all statements below are conditional on that guide. Write $\rho_m = \max_{k \ne m}\langle v_k, v_m\rangle < 1$.

Theorem 14 (Lobe-dictionary detection, localisation, and lower bound)**.** *For $Y = g + \epsilon$, $g \in \mathrm{range}(P)$ and $\epsilon \sim N(0, \sigma^2 I)$, the scan*

$$S = \max_{1 \le m \le M} \frac{\langle v_m, Y\rangle}{\sigma} \tag{29}$$

has level at most α at threshold $t_\alpha = \sqrt{2\log(M/\alpha)}$. Under $Y = g + \sigma\mu v_m + \epsilon$, its miss probability is at most β whenever

$$\mu \ge t_\alpha + \sqrt{2\log(1/\beta)}. \tag{30}$$

Moreover, the maximiser of the scan equals m with probability at least $1-\delta$ if

$$\mu(1 - \rho_m) > 2\sqrt{2\log(2M/\delta)}. \tag{31}$$

Conversely, if the v_m are orthonormal and $\alpha+\beta<1$, no level-α test has power greater than $1-\beta$ uniformly over all M alternatives when

$$e^{\mu^2}-1 \le 4M(1-\alpha-\beta)^2. \tag{32}$$

Under a uniform unknown index, every index estimator also has mean error at least $1-(\mu^2+\log 2)/\log M$. Thus both detection and localisation have critical standardised lobe norm of order $\sqrt{\log M}$. Over the stated orthonormal subclass, the scan therefore matches the lower-bound order $\sqrt{\log M}$, with an explicit constant gap; this is not a minimax claim for every correlated HCRD dictionary.

Proof. Under the null each scan coordinate is standard Gaussian, possibly correlated, so a Gaussian tail bound and union bound give $\Pr(S>t_\alpha)\le Me^{-t_\alpha^2/2}=\alpha$. Under the stated alternative, the true coordinate is $N(\mu,1)$, which proves the miss bound. With probability at least $1-\delta$, all M centred coordinates have magnitude at most $u=\sqrt{2\log(2M/\delta)}$. The true coordinate then exceeds every competitor when $\mu-u>\mu\rho_m+u$.

For the lower bound, restrict the affine nuisance to $g=0$ and mix the M orthogonal alternatives uniformly. Its chi-squared divergence from the null is $(e^{\mu^2}-1)/M$, hence total variation is at most one half of its square root. The displayed condition makes average power at most $1-\beta$, so at least one alternative has no greater power. Finally the pairwise Gaussian Kullback–Leibler divergence is μ^2; Fano's inequality [9] gives the localisation claim. □

The theorem concerns fixed or independently selected residual shapes. It does not make the discontinuous same-replicate knot selector innocuous, and its orthogonal lower-bound subclass is a benchmark rather than a claim that all HCRD lobes are mutually orthogonal.

The finite dictionary is not the mathematical end point: location, scale, and asymmetry are naturally continuous. Let Θ be compact, let w_θ be raw lobe samples, and suppose $\inf_\theta\|(I-P)w_\theta\|_2>0$. Define

$$v_\theta=\frac{(I-P)w_\theta}{\|(I-P)w_\theta\|_2}, \qquad d(\theta,\vartheta)=\|v_\theta-v_\vartheta\|_2, \tag{33}$$

and let $N(\epsilon)$ be the covering number in d. Write $J(\Theta)=\int_0^2\sqrt{\log N(\epsilon)}\,d\epsilon$.

Theorem 15 (Continuous chord-lobe scan)**.** *Assume $\theta\mapsto v_\theta$ is continuous, $J(\Theta)<\infty$, and the family is fixed or selected from an independent guide. For $Y=g+e$, $g\in\operatorname{range}(P)$ and $e\sim N(0,\sigma^2 I)$, the continuous scan*

$$S_c=\sup_{\theta\in\Theta}\frac{\langle v_\theta,Y\rangle}{\sigma} \tag{34}$$

has level at most α at the safe threshold

$$t_\alpha=24J(\Theta)+\sqrt{2\log(1/\alpha)}. \tag{35}$$

Under $Y=g+\sigma\mu v_{\theta_0}+e$, its miss probability is at most β whenever $\mu\ge t_\alpha+\Phi^{-1}(1-\beta)$.

If $G_\eta\subset\Theta$ is a finite η-net in d, scanning its M templates at $\sqrt{2\log(M/\alpha)}$ still has level at most α. Every continuous alternative has a net template with correlation at least $1-\eta^2/2$; hence the sieve has miss probability at most β whenever

$$\mu\ge\frac{\sqrt{2\log(M/\alpha)}+\Phi^{-1}(1-\beta)}{1-\eta^2/2}, \qquad \eta<\sqrt{2}. \tag{36}$$

Proof. Affine residualisation centres the Gaussian process $Z_\theta=\langle v_\theta,e\rangle/\sigma$; its canonical metric is exactly d and every marginal variance is one. Dudley's entropy inequality bounds $\mathbb{E}\sup_\theta Z_\theta$ by $24J(\Theta)$ [13]. Gaussian concentration about this mean gives the stated tail [39]. At the true template the alternative score is $N(\mu,1)$, proving the power statement. For the sieve, the finite union bound gives level and the unit-vector identity $\langle u,v\rangle=1-\|u-v\|_2^2/2$ gives the approximation factor. □

Theorem 16 (Rank-aware projected-norm control for continuous scans). *Let every v_θ be a unit vector in a fixed or independent-guide q-dimensional subspace $U \subseteq \operatorname{range}(I-P)$. If $Q_q(1-\alpha)$ is the $(1-\alpha)$ quantile of χ_q^2, set*

$$t_{q,\alpha} = \sqrt{Q_q(1-\alpha)}. \tag{37}$$

Then the same continuous scan has level at most α at $t_{q,\alpha}$. Its miss probability under $g + \sigma\mu v_{\theta_0}$ is at most β whenever

$$\mu \geq t_{q,\alpha} + \Phi^{-1}(1-\beta). \tag{38}$$

For a parameter metric d_Θ, define

$$\Delta_{\theta_0}(r) = \inf_{d_\Theta(\theta,\theta_0)\geq r} \{1 - \langle v_\theta, v_{\theta_0}\rangle\}. \tag{39}$$

If this modulus is positive, every measurable scan maximiser lies within r of θ_0 with probability at least $1-\delta$ whenever

$$\mu\Delta_{\theta_0}(r) > 2t_{q,\delta}. \tag{40}$$

A closed-form, dependency-free replacement for the projected-norm quantile is

$$\bar{t}_{q,\alpha} = \sqrt{q + 2\sqrt{q\log(1/\alpha)} + 2\log(1/\alpha)}. \tag{41}$$

Proof. Under the affine null, $S_c \leq \|\Pi_U e/\sigma\|_2$ by Cauchy–Schwarz, simultaneously over the uncountable unit-template family. Because U is fixed relative to the scoring noise, the squared projected norm is χ_q^2. Thus the χ_q quantile exactly calibrates the dominating projected norm and gives scan size at most α; it is the scan's exact null quantile only when the template family attains the relevant unit-sphere directions. The true-template coordinate is $N(\mu, 1)$, giving the stated sufficient power condition. On the projected-norm event at level δ, every signed noise inner product is bounded by $t_{q,\delta}$; the mean score gap outside radius r is at least $\mu\Delta_{\theta_0}(r)$, proving localisation. The last display follows from the Laurent–Massart chi-squared tail bound [26]. □

No entropy or discretisation is needed for this conservative envelope in Theorem 16; compact continuity supplies measurability. For affine residualisation on a strictly increasing grid, $U = \operatorname{range}(I-P)$ has $q = n-2$. Selecting U from the scoring noise, using unnormalised templates, or replacing σ by a same-sample estimate invalidates the displayed pivot.

The entropy input can itself be certified analytically. If $\Theta = \prod_{j=1}^d [a_j, b_j]$ and $\|v_\theta - v_\vartheta\|_2 \leq L\|\theta - \vartheta\|_2$, coordinate grids give

$$N(\epsilon) \leq \prod_{j=1}^d \left(1 + \frac{L\sqrt{d}\,(b_j - a_j)}{\epsilon}\right), \tag{42}$$

which has an explicit finite entropy integral.

Corollary 4 (Interior triangular-family certificate). *Let $x_0 < \cdots < x_{n-1}$ have maximum gap Δ. Parameterise a unit-height triangular lobe by centre c, support width s, and apex fraction a on a box whose supports lie strictly inside (x_0, x_{n-1}). With*

$$h = s_-\min\{a_-, 1-a_+\}, \quad \gamma = 1 - \frac{\Delta}{2h} > 0, \quad B = \sqrt{\frac{7}{2} + \max\{|a_- - 1/2|, |a_+ - 1/2|\}^2 + s_+^2}, \tag{43}$$

the residuals and normalized template map satisfy

$$\inf_\theta \|(I-P)w_\theta\|_2 \geq \frac{\gamma}{\sqrt{2}}, \qquad \|v_\theta - v_\vartheta\|_2 \leq \frac{4\sqrt{n}B}{h\gamma}\|\theta - \vartheta\|_2. \tag{44}$$

Thus Theorem 15 has a fully analytic certificate for this declared location–width–asymmetry family.

Proof. A grid point lies within $\Delta/2$ of every apex, so the sampled height is at least γ. Contrast that sample using a vector supported on it and the two zero domain-endpoint samples and orthogonal to 1 and x to obtain the residual lower bound. On each triangle branch, derivatives with respect to the left, apex, and right knots are bounded by $\sqrt{2}/h$. Combining their parameter Jacobian bound B, summing over samples, projecting nonexpansively, and applying the $2/m$ normalization inequality with $m = \gamma/\sqrt{2}$ gives the stated Lipschitz constant. □

The constants are deliberately safe. Boundary-touching supports, general polygonal lobes, estimated or dependent noise, and sharper generic chaining remain outside this corollary.

9 Replicate-split inference for selected structures

Let a guide G be independent of a scoring replicate $Y = f + e$, $e \sim N(0, \sigma^2 I)$. HCRD may select any finite family of intervals and shapes from G. On an active interval I_s, let c_s be the trapezoidal coefficient vector for signed area relative to the endpoint chord; it annihilates every affine vector. Alternatively let $q_s(G)$ be the complete guide detail and $v_s = (I - P_s)q_s(G)$, where P_s projects onto $\operatorname{span}\{\mathbf{1}, x\}$.

Theorem 17 (Exact post-selection area and shape pivots)**.** *Conditionally on G, every nondegenerate selected structure satisfies*

$$\frac{c_s^T Y_{I_s}}{\sigma \|c_s\|_2} \sim N\left(\frac{c_s^T f_{I_s}}{\sigma \|c_s\|_2}, 1\right), \qquad \frac{v_s^T Y_{I_s}}{\sigma \|v_s\|_2} \sim N\left(\frac{v_s^T f_{I_s}}{\sigma \|v_s\|_2}, 1\right). \tag{45}$$

Two-sided Gaussian p-values followed by Holm correction control conditional and unconditional family-wise error over all selected, overlapping structures. Under $f_{I_s} = a + bx + a_s v_s$, the matched-shape noncentrality is $a_s \|v_s\|_2/\sigma$.

Proof. Conditionally on the independent guide, c_s and v_s are fixed. Fixed linear functionals of the Gaussian scoring noise have the displayed laws, and both coefficient vectors annihilate affine means. Holm control requires valid marginal p-values but no independence among the overlapping tests. □

Reusing the scoring noise to select its largest structure invalidates these pivots; with two independent null candidates, selecting the smaller naive p-value already inflates rejection probability to $1 - (1 - \alpha)^2$. Independent replicates are therefore a substantive assumption, not a notation for adjacent samples in one autocorrelated series.

Theorem 18 (Buffered single-stream post-selection pivots)**.** *Let $Y_t = f_t + e_t$, where e is a mean-zero Gaussian m-dependent process. Partition the stream into contiguous scoring blocks B_k and let guide σ-field $\mathcal{G}_k$ use only indices at distance greater than m from B_k. Suppose a finite family of contrasts c_{ks}, supported on B_k, is $\mathcal{G}_k$-measurable and the positive block covariance Σ_{B_k} is known. Under $H_{ks} : c_{ks}^T f_{B_k} = 0$,*

$$Z_{ks} = \frac{c_{ks}^T Y_{B_k}}{\sqrt{c_{ks}^T \Sigma_{B_k} c_{ks}}} \,\Bigg|\, \mathcal{G}_k \sim N(0, 1). \tag{46}$$

Consequently two-sided Gaussian p-values remain valid after guide-side HCRD selection, and Holm correction over every fold and selected structure controls strong family-wise error without assuming cross-fold independence.

Proof. m-dependence and the buffer make e_{B_k} independent of $\mathcal{G}_k$. Conditional on that guide, each contrast is fixed; Gaussianity and the full block covariance give the displayed law. Holm requires valid marginal p-values, not mutual independence. □

Every index can be scored once: only the m neighbours of a scoring block are omitted from that fold's guide, not from their own scoring folds. The theorem is exact for known-covariance Gaussian finite-memory noise. An AR(1) process is not m-dependent for any finite m; estimated covariance, data-selected lag, and general mixing therefore require additional error control.

Theorem 19 (Nested subtree gatekeeping)**.** *Let T be a fixed or independent-guide finite rooted tree and associate to each node v the intersection null H_v that there is no signal anywhere in the subtree rooted at v. Let p_v be marginally valid for H_v under arbitrary dependence. The tree and all local level allocations are deterministic or measurable with respect to the independent guide. Assign $a_{\mathrm{root}} = \alpha$ and local levels satisfying*

$$\sum_{u \in \mathrm{child}(v)} a_u \leq a_v. \tag{47}$$

Test the root at a_{root} and a non-root node at a_v only after every ancestor rejects. Then the probability of rejecting any true null is at most α.

Proof. The minimal true nodes form an antichain, and rejection of any true descendant requires rejection of one such boundary node. Child-budget conservation implies inductively that the levels assigned to any antichain sum to at most α. A union bound over the minimal true nodes completes the proof. □

Allocating a parent's budget to children in proportion to descendant leaf counts conserves the full budget and retains level α along a one-child chain. This may improve power for coherent nested alternatives but does not uniformly dominate Holm for isolated leaves. Geometric containment alone is insufficient: the node nulls must be subtree intersection nulls and a same-scoring-noise tree does not automatically have valid node p-values.

10 Experiments

The primary comparison protocol was frozen locally before inspecting results; it was not deposited in a public preregistration registry. It tests affine chord baselines with alternating convex/concave equal-amplitude parabolic lobes. Generic smoother hyperparameters are selected by an oracle against the known latent baseline separately for each trial. The primary endpoint is baseline mean squared error. A task-specific superiority statement requires a paired bootstrap interval below zero and a multiplicity-corrected exact sign test. Out-of-class tests include polynomial trends, crossing chirps, weak oscillations under strong background curvature, boundary outliers, irregular sampling, and non-Gaussian noise.

The theorem-linked R1 experiment varied $\rho = \gamma/\tau_\delta$ over 11 values on four fixed event-count/width configurations: $(J, m) = (2, 4), (4, 8), (8, 16), (16, 8)$. Each cell contained 1000 independent draws, for 44,000 trials in total. Figure 2 shows a clear transition. Exact recovery ranged from 0.433 to 0.879 across configurations at $\rho = 1.45$, from 0.980 to 0.991 at $\rho = 1.75$, and was 0.999 throughout at $\rho = 1.90$. In all eight cells above the strict theorem boundary $\rho > 2$, it was 1.000; the smallest cellwise 95% Wilson lower bound was 0.9962. The joint event of exact knots and both reconstruction bounds had minimum probability 0.9980 above the boundary (minimum Wilson lower bound 0.9927). The result validates a sufficient condition on its declared class, not a necessary boundary for arbitrary waveforms.

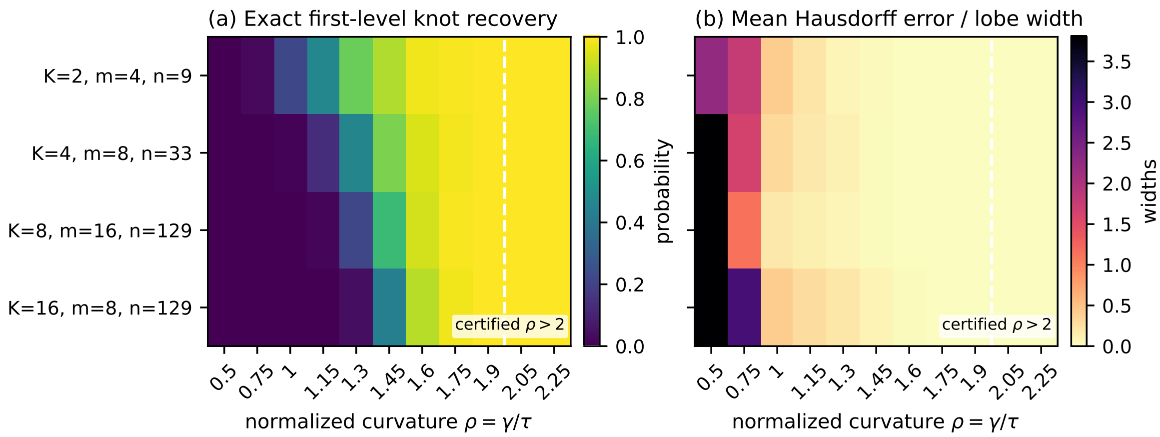

Figure 2: Theorem-linked recovery phase diagram under iid Gaussian noise. Rows vary lobe count J, samples per lobe m, and total length n; columns vary normalized active curvature $\rho = \gamma/\tau_\delta$. Left: exact first-level knot recovery. Right: symmetric Hausdorff boundary error divided by lobe width. The dashed line marks the strict sufficient region $\rho > 2$; all 8000 draws in that region recovered the exact knots.

Table 2: Mean baseline MSE over 100 preregistered trials. Oracle labels indicate per-signal ground-truth tuning.

Method	Exact, $\sigma = 0$	Exact, $\sigma = 0.1$	Variable, $\sigma = 0.1$
Centred HCRD	0	0.787	0.887
Thresholded HCRD	3.78e-32	0.00778	0.312
Adaptive guided HCRD	0.0124	0.0385	0.0856
Gaussian oracle	0.0433	0.0382	0.0632
Moving-average oracle	0.0516	0.0486	0.0743

As a computational audit rather than a proof, we also simulated 50,000 null and alternative draws for a fixed 24-template compact-lobe dictionary in 257 samples. Its maximum coherence was 0.0376. At the conservative theorem threshold for $\alpha = 0.05$, empirical false rejection was 0.00524 (Wilson interval $[0.00464, 0.00591]$). At the sufficient power norm, miss probability was 0.0363 versus the guaranteed bound 0.20; at the localisation norm, no error was observed (upper Wilson limit 7.68×10^{-5}) versus the bound 0.10. These checks expose conservatism and implementation errors; they do not replace Theorem 14.

We separately audited Theorem 15 on 129 samples and a 595-template centre–width–asymmetry sieve. In 50,000 null draws the finite threshold 4.332 was exceeded 22 times: empirical FWER 0.00044 with 95% Wilson interval $[0.000276, 0.000666]$. Its true-template Gaussian sufficient norm 5.1739 for $\beta = 0.2$ gave empirical power 0.8773 in 10,000 draws. For the complete continuum, $q = 127$: the projected-norm envelope threshold 12.4218 was exceeded in 2537/50,000 direct projected-norm draws, giving 0.05074 ($[0.04883, 0.05270]$), while the closed-form Laurent–Massart threshold 13.1150 was exceeded with probability 0.00486. The resulting conservative 80%-power sufficient norm is 13.2634. This replaces the earlier Dudley sufficient norm 285.1103 without treating the numerical grid as the continuum. These draws audit the dominating norm, not the generally smaller scan null. The finite sieve remains sharper because it declares only 595 templates rather than the full residual subspace.

On 100 fresh noiseless exact-class trials, centred HCRD recovered the latent baseline to numerical zero. It had lower MSE in every trial than the historical legacy rule and the oracle-tuned Gaussian, moving-average, and Fourier comparators. All corresponding paired bootstrap intervals were below zero and Holm-adjusted sign-test values were below 1.6×10^{-29}. Thresholded HCRD and an endpoint-only RDP baseline were also exact within 10^{-12} and are reported as ties, not losses. That RDP solution uses only the two domain endpoints: it recovers the global affine baseline but not the $J-1$ lobe boundaries, so it does not tie the boundary-recovery endpoint in Figure 2. On the exploratory variable-amplitude, piecewise-affine class without noise, centred HCRD had mean MSE 0.0179, versus 0.0645 for the Gaussian oracle, but the raw method failed under sample-level white noise. Adaptive guidance improved knot accuracy substantially; its pilot-calibrated scale is not yet supported by a distribution-free theorem.

An external-method experiment then used 50 further exact-class trials. On the noiseless condition, centred HCRD had lower baseline MSE in all 50 trials than the final EMD residue, the lowest-frequency VMD mode, and per-signal oracle-tuned ℓ_1 trend filtering. The respective mean competitor MSEs were 0.0108, 0.511, and 0.0425, versus zero for HCRD; every paired bootstrap interval was below zero and the Holm-adjusted sign-test value was 1.07×10^{-14}. A supplementary direct comparison on the same 50 fixed draws used the final ITD baseline returned by PySDKit 0.4.54. ITD mean MSE was 0.1473; the HCRD–ITD paired difference was -0.1473 ($[-0.1871, -0.1085]$), with 50/50 HCRD wins and Holm-adjusted exact sign $p = 1.24 \times 10^{-14}$. EMD, VMD, and ITD settings follow their standard algorithms [19, 12, 17]; oracle tuning deliberately favours the smoothing comparators. This confirmation is specific to the chord-lobe class, rather than evidence of global dominance over ITD's time–frequency use cases. Although DPT is an exact nonlinear pulse decomposition, its native components are constant pulses and do not define a unique final chord baseline at a matched scale; it is therefore included in the structural comparison rather than assigned an artificial baseline-MSE endpoint.

After that result, two fresh 100-trial noisy confirmations were prospectively frozen for the plug-in MAD-thresholded variant. Table 3 reports their means. At $\sigma = 0.03$, its win rates against EMD, VMD, ℓ_1, and Gaussian were respectively 0.86, 0.99, 0.99, and 0.98; at $\sigma = 0.10$ they were 0.81, 0.93, 0.96, and 0.74. All paired bootstrap intervals remained below zero. The largest Holm-adjusted sign-test value was 1.67×10^{-6}. This finite-sample confirmation concerns the specified generator and plug-in estimator; Theorem 9 itself assumes known or externally estimated noise scale.

Table 3: Mean baseline MSE for external comparisons. The noiseless column is E1 (50 trials); noisy columns are fresh C2 confirmations (100 trials each).

Method	$\sigma = 0$	$\sigma = 0.03$	$\sigma = 0.10$
Centred HCRD	0	0.762	0.789
Thresholded HCRD	2.92×10^{-32}	0.00356	0.0195
EMD residue	0.0108	0.0288	0.0531
VMD low mode	0.511	0.0344	0.0396
ℓ_1 trend oracle	0.0425	0.0419	0.0443
Gaussian oracle	0.0378	0.0336	0.0383

Two additional external decompositions were then tested under independently frozen, fresh-seed protocols. Improved complete ensemble EMD with adaptive noise (CEEMDAN) [8] used 20 noise realizations and an oracle selected, separately for every signal, the cumulative one-to-four slowest components that minimized latent-baseline MSE. Fast Iterative Filtering used the author-linked Python implementation with its published defaults [6]. Its oracle used the analogous cumulative slow-tail selection. Both oracles use unavailable ground truth and therefore favour the comparators.

Table 4 reports the results. In E2, HCRD won 50/50, 48/50, and 35/50 paired trials at noise standard deviations 0, 0.03, and 0.10; the largest Holm-adjusted sign-test value was 0.00660, and all three 95% bootstrap intervals for HCRD minus CEEMDAN lay below zero. In E3 the corresponding wins over oracle Iterative Filtering were 50/50, 50/50, and 49/50, with largest adjusted value 9.06×10^{-14} and again all intervals below zero. These confirmations strengthen only the stated chord-lobe baseline task: CEEMDAN and Iterative Filtering target oscillatory mode decompositions rather than endpoint-chord recovery.

Table 4: Fresh modern-decomposition confirmations (50 signals per condition). The HCRD reference is centred at $\sigma = 0$ and MAD-thresholded otherwise. E2 and E3 use independent seeds and must be read as paired rows within study.

Study	Method	$\sigma = 0$	$\sigma = 0.03$	$\sigma = 0.10$
E2	HCRD reference	0	0.000400	0.00636
E2	CEEMDAN oracle slow tail	0.0135	0.0137	0.0312
E3	HCRD reference	0	0.000398	0.0124
E3	Iterative Filtering oracle tail	0.135	0.156	0.143

Finally, an exploratory Case Western Reserve University bearing study used 12 kHz drive-end recordings for four states and four motor loads [37]. Complete recordings, rather than randomly mixed windows, defined leave-one-load-out folds. Raw summary features already reached mean balanced accuracy 0.982; Gaussian, wavelet, EMD, and raw HCRD representations all reached 1.000. Thus the data exhibit a ceiling effect and provide no evidence that HCRD is superior. Guided HCRD was worse at 0.945, including 0.781 on the 3-hp fold. The experiment therefore identifies a ceiling regime in which HCRD provides no classification benefit.

Runtime protocol P1 processed the same 384 CWRU windows five times per mode in randomized order after a whole-system load gate; compiled-library thread counts were fixed to one, and pool startup and serialization were timed. Every one of the 35 feature matrices was bitwise identical to the saved R1 matrix. On an Intel Core Ultra 7 155H, serial median latency was 10.279 s. Two, four, eight, and sixteen worker processes achieved respectively 1.76, 2.16, 2.47, and 2.00 times speedup; four and eight Python threads achieved only 0.55 and 0.60 times serial performance because the reference knot walk is GIL-bound. The complete machine-specific scaling results are provided in the repository. Levels within one signal remain sequential, whereas independent windows, records, and channels are exactly parallel.

We implemented Corollary 1 by storing only nested knot sets and materializing dense arrays on demand. Protocol P2R required five consecutive one-second whole-system samples, each at most 20%, before every trial. Every timed hierarchy had the same canonical knot digest, and materialized features for all 384 windows were bitwise identical to R1. Sparse serial construction reduced the median from 14.461 to 1.217 s, an 11.89-fold representation-level speedup; four processes further reduced it to 0.976 s (1.25-fold versus sparse serial). Eight processes did not amortize startup and serialization on this short batch.

Protocol P3 therefore froze a tenfold larger 3840-window throughput batch under the same per-trial load gate. Table 5 shows monotone process scaling to a 1.96-fold end-to-end speedup with eight workers. This supports exact independent-signal parallelism, but not ideal linear scaling; all timings include pool construction, serialization, output collection, and hierarchy allocation.

Table 5: Strictly load-gated sparse runtime results. Speedup is relative to dense serial in P2R and sparse serial in P3.

Protocol	Representation/backend	Workers	Signals	Median (s)	Speedup
P2R	dense serial	1	384	14.461	1.00
P2R	sparse serial	1	384	1.217	11.89
P2R	sparse process	4	384	0.976	14.82
P3	sparse serial	1	3840	13.682	1.00
P3	sparse process	2	3840	11.911	1.15
P3	sparse process	4	3840	8.669	1.58
P3	sparse process	8	3840	6.963	1.96

A separate new-seed confirmation assessed the quadratic proximal guide with a single pilot-selected but then frozen value $\lambda = 3$ on the variable-amplitude chord-lobe generator. Thirty independent latent signals, each with ten noise repetitions, were used at every noise level; inference averaged repetitions within latent signal. Table 6 shows a morphology advantage. Against the adaptive Gaussian guide, paired F1 differences were 0.274, 0.244, and 0.191 for $\sigma = 0.01, 0.03, 0.05$, with all 95% bootstrap intervals strictly positive; at $\sigma = 0.10$ the difference was 0.011 with interval $[-0.015, 0.039]$, an explicit tie. Quadratic guidance also exceeded raw, thresholded, and fixed-Gaussian HCRD at every noise level after Holm correction. The maximum sampled guide perturbation ratio was 0.366, consistent with Theorem 7; after hard HCRD it reached 2.543, confirming that the global theorem must not be transferred to the knot output.

Table 6: Frozen S2 confirmation. F1 matches latent chord knots within one sample. MSE is against the latent baseline.

σ	Quad. F1	Adaptive F1	Fixed-Gauss. F1	Quad. MSE	Adaptive MSE
0.01	0.716	0.442	0.410	0.0730	0.0722
0.03	0.720	0.476	0.399	0.0732	0.0863
0.05	0.722	0.531	0.338	0.0732	0.0922
0.10	0.720	0.709	0.230	0.0727	0.0964

10.1 Expert-labelled chromatographic lobes

We evaluate HCRD for LC–MS peak-shape curation. A genuine extracted-ion chromatogram (EIC) is approximately a compact, predominantly one-sign lobe over a local baseline, whereas shoulders, split peaks, baseline fragments, and noise violate that structure. The public data of Müller et al. [31] contain 255,000 EICs: 5000 candidate ions evaluated in each of 51 real samples by mass- spectrometry experts. Before reading the arrays or classification table, we froze a double group holdout. Sample identifiers and candidate-ion identifiers were hashed independently; a case was admitted only when both hashes assigned it to the same train, validation, or confirmation block. Confirmation thus shared neither samples nor candidate ions with model development.

Every representation received both time windows shown to the experts. The same fixed histogram-gradient-boosted classifier was trained on robustly resampled raw EICs, a conventional multiscale morphology bank, raw plus one HCRD level, raw plus the complete eight-level HCRD hierarchy, HCRD scalar geometry without waveform channels, or polygon-area/energy summaries alone. Validation selected the conventional domain bank before the confirmation block was unlocked. Primary inference used a 10,000-replicate two-way cluster bootstrap over both held-out samples and held-out candidate ions.

Table 7 shows the locked result. Eight-level HCRD achieved the largest confirmation AP, 0.4092 versus 0.3820 for the frozen comparator, and exceeded the one-level representation by

0.0321. The primary difference was 0.0272, but its cluster interval $[-0.0566, 0.1010]$ crossed zero because the double holdout left only eight independent confirmation samples. The prospective success rule therefore failed. The ablations are nevertheless informative: area alone was weakest, and scalar geometry without the signed detail channels was also below the conventional comparator. The useful object is the complete multilevel decomposition in combination with the raw shape; polygon mass is auxiliary rather than the method.

Table 7: Locked E1 expert-labelled LC–MS confirmation (2692 unambiguous EICs, 297 peaks; eight sample and 706 candidate-ion groups). AP is the primary threshold-free endpoint.

Representation	AP	ROC AUC	MCC	Balanced acc.
Raw EIC	0.3626	0.8714	0.4150	0.7782
Conventional domain bank	0.3820	0.8801	0.4379	0.7866
Raw + HCRD level 1	0.3771	0.8696	0.4026	0.7594
Raw + HCRD levels 1–8	**0.4092**	**0.8844**	**0.4418**	**0.8133**
HCRD scalar geometry	0.3345	0.8415	0.3479	0.7409
Polygon area/energy only	0.3151	0.8254	0.3369	0.7451

The independent E2 study then used the two fully expert-labelled HILIC studies of Kumler et al. [24]. Their two-parameter logistic quality score combines correlation with an idealized beta-shaped peak and a residual SNR. The source study found this simple model unusually robust under Falkor/MESOSCOPE transfer, making it a stronger and more interpretable primary comparator than a broad post-hoc model search. We used an independent global-window reimplementation of the published two-component score from the raw mzML feature boxes; on 1495 Falkor features its Pearson correlations with the authors' SNR and bell-correlation values were 0.906 and 0.789 (Spearman 0.871 and 0.805).

Before reading any raw waveform, we froze both transfer directions, Good versus Bad labels, exclusion of Ambiguous/standards-only cases, per-file robust aggregation, one identical standardized logistic learner, and 10,000 paired feature-bootstrap replicates. The classifier was fit on all 1821 unambiguous Falkor features and applied without refitting to 2193 MESOSCOPE features, then the direction was reversed. Figure 3 reports the result. HCRD-8 plus qscore improved AP over qscore from 0.7774 to 0.8955 in Falkor-to-MESOSCOPE and from 0.7984 to 0.8990 in the reverse direction. The paired differences were 0.1181 ($[0.0613, 0.1781]$) and 0.1005 ($[0.0559, 0.1470]$); both Holm-adjusted values were 0.00040. Every prospective success component passed.

The waveform representations were not equal in width: after fixed per-file aggregation, qscore had 2 variables, the conventional domain bank 336, HCRD-1 534, and HCRD-8 2847. We therefore added a supplementary matched-capacity control, frozen before its scores were read, that expands the same 75 raw waveform inputs into Gaussian smoothing, first-derivative, and curvature channels and aggregates them to exactly 2847 variables. Its result is reported separately below because it was not part of the prospective E2 family.

Multilevel HCRD also exceeded one-level HCRD in both directions, by 0.0439 and 0.0544 AP; the second interval $[0.0076, 0.0955]$ remained positive after Holm correction ($p = 0.0372$), while the first $[-0.0167, 0.1060]$ did not. The best point representation was HCRD scalar geometry plus qscore (AP 0.9150) in the first direction and the multilevel area/energy spectrum plus qscore (0.9412) in the second. Conversely, full HCRD-8 did not establish a difference from the larger conventional domain bank: its two intervals were $[-0.0469, 0.0561]$ and $[-0.0479, 0.0154]$. Thus E2 confirms incremental multilevel convex-lobe information over the established minimal score, while the optimal compression of that hierarchy and superiority to every non-neural domain representation remain open.

The E2 bootstrap estimand is conditional on the fitted source-study model: it resamples held-out target features and compares predictions from the already trained learners. It therefore does not propagate source-model refitting or compound/adduct, m/z–retention-time-neighbourhood, and acquisition-file clustering. The intervals quantify target-feature uncertainty under this transfer protocol, not the uncertainty of repeating the complete experimental pipeline.

In the matched-capacity sensitivity analysis, the Gaussian-derivative control had AP 0.9063 versus 0.8955 for HCRD-8 in Falkor-to-MESOSCOPE; the HCRD minus control difference was -0.0108 ($[-0.0594, 0.0405]$, Holm-adjusted $p = 0.7653$). In the reverse transfer, HCRD-8 had AP 0.8990 versus 0.7815, a difference of 0.1175 ($[0.0624, 0.1654]$, adjusted $p = 0.00040$). Thus raw dimensionality does not explain the reverse-direction gain, but the first direction provides no evidence that HCRD uniformly dominates a comparably rich conventional scale-space bank.

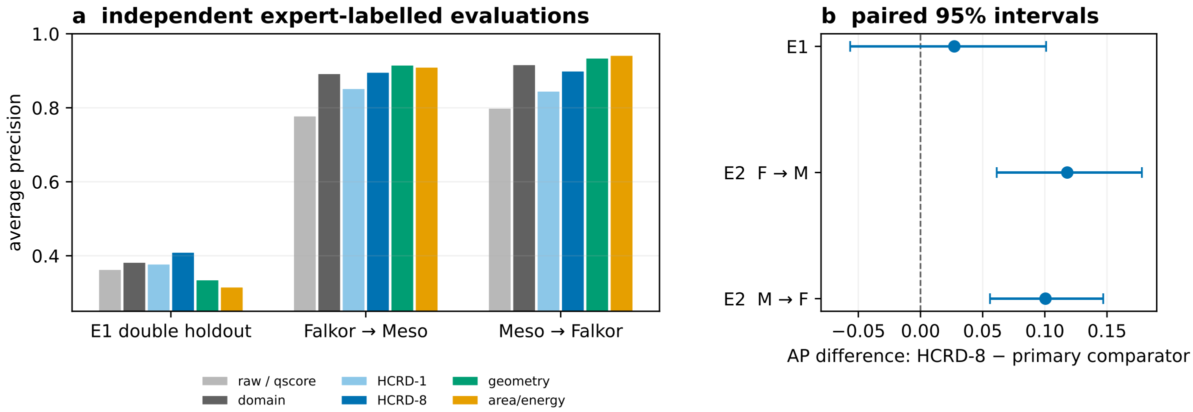

Figure 3: Independent expert-labelled LC–MS evidence. Panel a shows average precision for the first double holdout and both no-refit E2 transfers; "raw / qscore" denotes the task-appropriate minimal representation. Panel b shows the predeclared HCRD-8 difference from the primary comparator. The E1 interval resamples both sample and candidate-ion groups; E2 target-feature intervals are conditional on the trained source models. F and M abbreviate Falkor and MESOSCOPE.

LCMS-E5 tested a different operational consequence on Pttime. The source authors manually labelled only 400 of 7781 features: those already assigned a probability above 0.9 by their two-variable model. We therefore froze E5 as a conditional residual-error reranking test, not as a third fully labelled cohort. Before any target waveform was inspected, HCRD-8, the learner, AP-bad endpoint, and success rule were fixed. The Falkor and MESOSCOPE cases were pooled for training and the model was applied without refitting to the 365 unambiguous Pttime cases (348 Good, 17 Bad). Archive inspection before feature extraction revealed both ion modes; the source pipeline's POS filename schema fixed the compatible 52 positive-mode files, and this correction was recorded before any E5 score was computed.

Table 8 shows that HCRD-8 passed both prospective conditions. It attained AP-bad 0.5085 versus 0.0391 for qscore, a difference of 0.4694 with a paired class-stratified bootstrap interval $[0.2489, 0.6762]$ ($p = 0.00020$), and exceeded HCRD-1 by 0.2240 ($[0.0707, 0.3544]$, $p = 0.00060$). Seven of the 17 residual bad features were placed in the top 19 (5%) review positions, versus none under qscore. At 1%, 5%, and 10% queue budgets (4, 19, and 37 reviews), HCRD-8 retrieved 4, 7, and 8 bad cases, respectively, versus 0, 0, and 1 for qscore. Hence its 5%-budget precision was 36.8% and residual-bad recall was 41.2%. To reach at least 50% recall (9/17), HCRD-8 required 44 reviews versus 284 under qscore, an 84.5% workload reduction at the same integer recall target. HCRD-8 also had the best point estimate, but its 0.1212 AP advantage over DOMAIN+Q was not established ($[-0.0100, 0.2794]$, $p = 0.0818$). Thus E5 confirms practical conditional triage value and the need for multiple levels; selection of the labelled population prevents claims about all 7781 Pttime features, recall, calibration, or FDR.

Table 8: Frozen E5 Pttime residual-bad reranking within the 365 unambiguous source-model-selected features. AP-bad is primary; queue entries give bad features retrieved among 17 at review budgets of 4, 19, and 37 cases.

Representation	AP-bad	ROC AUC-bad	1%	5%	10%
qscore	0.0391	0.3403	0	0	1
DOMAIN+Q	0.3873	0.8115	4	6	6
HCRD-1+Q	0.2845	0.7792	2	5	7
HCRD-8+Q	**0.5085**	**0.8646**	**4**	**7**	**8**
HCRD geometry+Q	0.3540	0.8081	3	6	7
Area/energy+Q	0.1367	0.7194	1	3	5

E6 then tested a more severe external shift using the manually rated TARDIS FAME saliva panel [41]. The protocol, primary cohort, representations, pooled Falkor+MESOSCOPE learner, AP-bad endpoint, two co-primary comparisons, 20,000-replicate paired bootstrap, and seed were frozen before any rating row was inspected. The 5.67-GB Zenodo archive matched its published MD5. A pre-score availability audit found 119 positive-polarity QC mzXML files but no numerical negative-polarity EICs; the positive stratum was therefore evaluated and the negative stratum declared unavailable rather than approximated from diagnostic images. Fixed 10-ppm, 60-s boxes yielded valid feature banks for all 212 keyed positive targets, with median QC availability 0.9412; 108 Good and 84 Bad targets entered the binary endpoint and 20 Ambiguous targets only the ordered-label sensitivity analysis.

Table 9 reports the no-refit result. HCRD-8+Q improved AP-bad over qscore by 0.0725, but the percentile interval $[-0.0155, 0.1345]$ crossed zero ($p = 0.1115$), so the first co-primary rule failed. It was tied with HCRD-1+Q (difference -0.0016, $[-0.0500, 0.0453]$), so the second also failed. DOMAIN+Q had the largest AP point estimate, while HCRD geometry+Q had the largest ROC AUC. Thus decomposition-derived shape information transferred in point estimate beyond qscore, but E6 does not establish that eight levels are needed or that HCRD beats a compact conventional domain bank under this acquisition and processing shift.

Table 9: Frozen E6 no-refit transfer to 192 unambiguous positive-polarity TARDIS/FAME targets. AP-bad is primary; enrichment is the Bad fraction in the top score decile divided by overall Bad prevalence.

Representation	AP-bad	ROC AUC-bad	Top-decile enrich.
qscore	0.4353	0.5230	0.686
DOMAIN+Q	**0.5193**	0.6030	1.257
HCRD-1+Q	0.5094	0.5902	**1.486**
HCRD-8+Q	0.5078	0.5998	1.371
HCRD geometry+Q	0.5044	**0.6049**	**1.486**
Area/energy+Q	0.3847	0.4377	0.571

Two additional tests evaluated the application boundary. LCMS-E3 used the public TargetedMSQC CSF panel and leave-one-run-out transfer across four runs [15]. This is a deliberately morphology-saturated setting: its 52 official variables already encode jaggedness, symmetry, modality, width, elution shift, transition ratios, and light/heavy consistency. On 576 eligible labelled transition pairs, QC52 AP was 0.9247 versus 0.8862 after adding the complete HCRD-8 bank. The difference was -0.0384 with a two-way run/peptide cluster interval $[-0.1455, 0.0505]$, and HCRD won only one of four held-out runs. HCRD-8 did exceed HCRD-1, but the prospective test failed. A later compact-geometry search reached AP 0.9372 and is reported only as development because it used these outcomes.

LCMS-E4 tested baseline recovery on 60 low-affine-deviation TIC windows from 20 raw NeatMS Dataset 1 mzML files [18]. Single and nested positive lobes were injected after the protocol and baselines were scored against their known real-background target. For nested lobes, HCRD-8 MSE was 0.1369, versus 0.1141 for per-case oracle asymmetric least squares and 0.1270 for oracle morphological opening. It nevertheless beat HCRD-1, Gaussian, Savitzky–Golay, and wavelet comparators on all 60 cases; the HCRD-8–HCRD-1 mean difference was -0.3942 with simultaneous interval $[-0.4082, -0.3801]$. A post-protocol mixed-polarity analysis remained worse than signed AsLS and symmetric morphology. Table 10 summarizes these task-specific boundaries.

Table 10: LC–MS application-boundary tests. Lower MSE is better in E4.

Test	HCRD	Comparator	Boundary
E3 TargetedMSQC	AP 0.8862	QC52 AP 0.9247	morphology-saturated features
E4 real background	MSE 0.1369	AsLS MSE 0.1141	sign-aware lower envelope

10.2 Cross-domain boundary searches

The hierarchy was also tested as a learned pulse-event representation and as a training-free transient score. Table 11 summarizes the results; full protocols, folds, plots, bootstrap intervals, and ablations are provided in the repository.

Table 11: Cross-domain boundary searches. Each row reports the corresponding held-out or cross-fitted comparison; none is pooled with the LC–MS evidence.

Test	Main result	Interpretation
CWRU bearings [37]	Raw features 0.982 balanced accuracy; several transforms, including HCRD, tied at 1.000	Ceiling effect; no HCRD superiority
QTDB QRS [25, 21]	Quadratic HCRD 22.74 ms; `ecgpuwave` 13.94/15.75 ms	Stable guide improves raw HCRD, but the specialist delineator is better
PPG-DaLiA [34]	Geometry F1 0.7945, mass-only 0.7659, tuned `find_peaks` 0.8013	Full hierarchy carries information beyond area, without detector superiority
TSB-AD/Yahoo [30, 32]	HCRD 0.5285 AUC–PR versus NORMA 0.3383; CNN 0.9052	Positive frozen replication, but only within the synthetic Yahoo family
Real WSD/AIOps [28]	HCRD 0.3756 versus POLY 0.4115; hybrid 0.3675 versus raw spectral residual 0.3850	No independent real-KPI confirmation

For pulse data, the full multilevel geometry significantly exceeded the area/energy-only ablation, confirming that polygon mass is auxiliary rather than the decomposition. For temporal anomaly detection we used $S(t) = \max_{1\leq \ell\leq 8} z_\ell(t)$, where z_ℓ is robust positive surprise in the levelwise area-density series. Its frozen Yahoo success did not transfer to real WSD or AIOps KPIs and therefore does not establish general anomaly-detection performance.

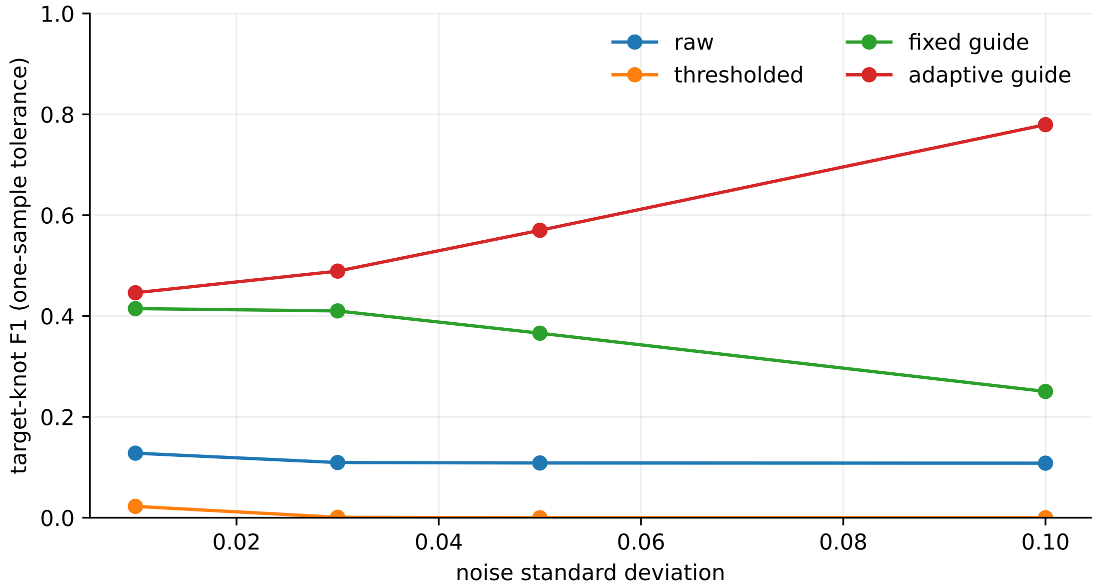

Figure 4: Knot accuracy under repeated Gaussian perturbations. Stability alone is insufficient: the thresholded method is stable because it often retains only endpoints, whereas adaptive guidance retains a compact set of meaningful knots.

Theorem 8 was audited on 1000 seeded perturbations, split equally between uniform and irregular grids. No numerical violation occurred; the maximum observed ratio $d_{\mathrm{SC}}/(C_x\|e\|_\infty)$ was 0.9899. On the four-sample discontinuity sequence, hard-baseline amplification grew from 20.5 to approximately 2×10^8 as the curvature scale decreased from 10^{-1} to 10^{-8}, while the persistence distance remained exactly one half of its theorem bound up to floating-point error (Figure 5).

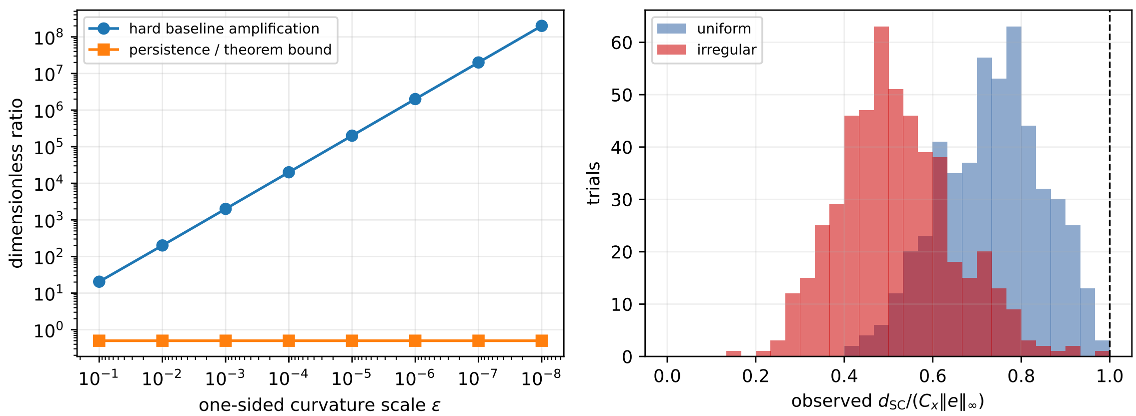

Figure 5: Hard-decision discontinuity versus signed-curvature persistence. Left: the hard baseline has unbounded amplification while the persistence metric remains within its global bound. Right: all 1000 uniform/irregular-grid checks remain below the theorem limit (dashed line).

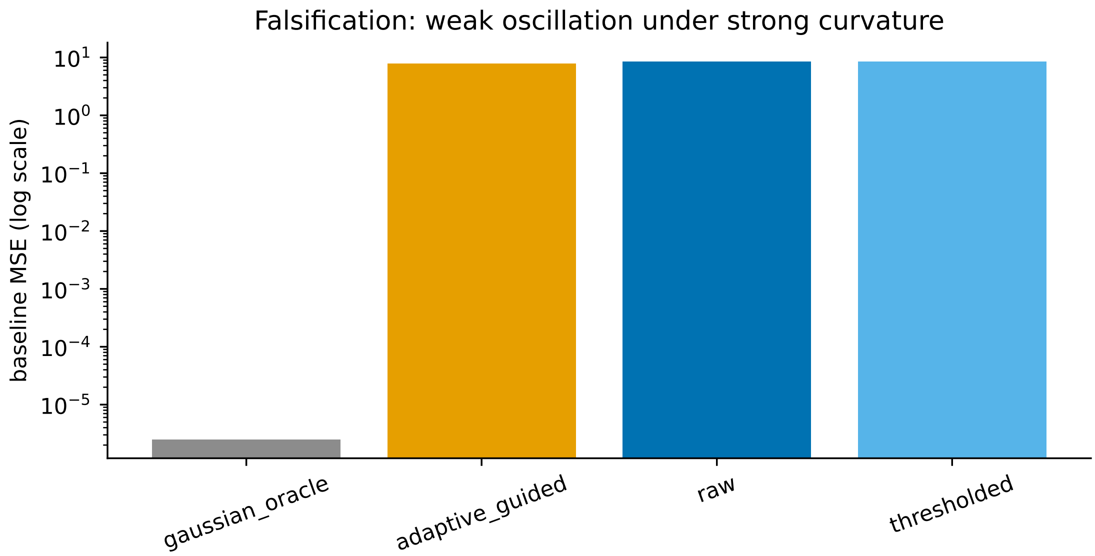

Figure 6: Limitation under strong background curvature. When background curvature exceeds the oscillatory curvature, raw HCRD cannot expose the oscillation and its chord baseline is inappropriate.

11 Discussion

HCRD occupies a specific rather than universal signal-processing niche. Its algorithmic appeal is its economy: divided-slope comparisons select maximal one-sign runs, endpoint chords form the next baseline, and subtraction gives an exactly reconstructing hierarchy. The resulting lobes expose support, sign, amplitude, residual shape, area, and level without a learned dictionary or a continuous smoothing parameter. This combination of simplicity and interpretability is the main contribution; its intended use is as a geometric representation of bounded waveform morphology, not as a replacement for every task-specific detector or decomposition.

The generative recovery result makes that niche testable. On alternating one-sign curvature blocks separated by sampled zero-curvature joins, the same geometry used by the algorithm yields exact first-level boundary recovery and explicit component-error control once $\gamma > 2\tau_\delta$. The R1 phase diagram tracks this margin across event count and width rather than showing a single favourable waveform. Its conservative boundary was respected in all 8000 above-threshold draws, while the empirical transition occurred slightly earlier. This is a class-specific mechanism result, not a separation theorem against every adaptive representation.

The clearest practical evidence is expert curation of pre-bounded LC–MS features. In the two frozen no-refit E2 transfers, multilevel HCRD improved average precision over an independent reimplementation of the published two-component score by 0.1181 and 0.1005, with Holm-adjusted $p = 0.00040$ in both directions. HCRD-8 had positive point differences from the one-level variant in both transfers, with multiplicity-adjusted support in one. The equal-dimensional conventional control tied HCRD-8 in one transfer, whereas HCRD-8 was superior in the reverse. E5 further supports residual-error reranking when only conditional labels are available. E6, however, did not confirm superiority under the larger TARDIS/FAME laboratory shift: the gain over qscore was 0.0725 only in point estimate, HCRD-8 tied HCRD-1, and a compact conventional feature bank remained best. Thus the evidence supports transferable value for pre-bounded compact-lobe curation, with hierarchy depth still dependent on the acquisition domain.

Results outside pre-bounded compact-lobe curation sharpen that positioning. HCRD added no benefit after a morphology-saturated LC–MS feature bank (E3), and sign-specific lower-envelope estimators were better chromatographic baseline models (E4). The ECG-specific `ecgpuwave` system retained lower QTDB boundary error; tuned `find_peaks` remained better overall on wearable PPG; and the supervised UCR, real WSD/AIOps anomaly, and bearing-RUL studies did not establish an advantage. These are important boundaries rather than a contradiction: a small generic geometry should not be expected to dominate specialist multichannel morphology rules, correct sign priors, or large supervised representations on their native objectives.

The empirical and theoretical results together suggest a practical selection rule. HCRD is most plausible when the object is already bounded, an endpoint chord is a meaningful local reference, and scientifically relevant structure appears as signed convex or concave excursions whose complete hierarchy may matter. It is less well matched to frequency separation, instantaneous frequency estimation, weak oscillations on a strongly curved trend, or noisy samplewise decisions without thresholding. Polygon mass and its levelwise time series are inexpensive auxiliary summaries, but the PPG and RUL evidence does not support replacing the full decomposition by area alone.

The simplicity also has computational consequences. With sparse retained knots and the halving property, total hierarchy work is linear in signal length; dense materialization can remain $O(n \log n)$. On P2R the knot-only implementation was 11.89-fold faster than dense materialization while preserving knots and downstream features bitwise. Independent signals admit exact process parallelism, including a 1.96-fold median speedup on the 3840-signal sparse P3 batch. Absolute speedups remain hardware- and batch-dependent, and the present Python implementation does not establish an optimal systems realization.

The theory now states the same boundary precisely. A hard knot map cannot be globally continuous while preserving every generic raw branch decision (Theorem 6); persistence matching and the proximal outer split provide well-posed stable alternatives, while deterministic and stochastic margins certify local hard-HCRD decisions. The explicit chord-lobe theorem turns those decision margins into boundary and component recovery on a declared generative class. The rank-aware scan, buffered m-dependent split, matched-replicate statistic, and nested-tree procedure supply finite-sample inference under explicit assumptions rather than broad asymptotics. These results justify a signed curvature-lobe representation, but do not identify its details with intrinsic modes.

The remaining gaps are consequently focused. The theory still assumes known or conservatively supplied scale, covariance, dependence lag, and stability radius; estimated versions, infinite-memory noise, unequal or unsampled lobe joins, and general polygonal templates remain open. A minimax separation from all competing adaptive dictionaries is not claimed. Empirically, representation selection must be frozen outside E2, and a further waveform-backed cross-laboratory LC–MS cohort is needed to resolve the roughly 0.07 AP effect seen in E6 and determine when multilevel depth transfers. The present evidence therefore supports a task-specific method with unusually transparent mechanics and failure modes. Broader state-of-the-art performance requires future independent confirmation.

Code and data availability

Implementation, tests, protocols, compact outputs, hashes, and environment instructions are at `https://github.com/Safreliy/hcrd`. Large derived caches are rebuilt locally.

A Boundary cases and detailed proof edges

This appendix expands the combinatorial and stability edges most vulnerable to off-by-one or hidden-decision assumptions. All indices below are local indices in the current eligible knot list; mapping them back to original sample indices preserves order and nesting.

Lemma 1 (Progress of a nonterminal centred interval)**.** *Suppose the current interval starts at eligible index r and the centred walk closes it at t before the terminal endpoint. Then $t - r \geq 2$.*

Proof. Let $p \geq r+1$ be the first active curvature after any insignificant entries. A nonterminal closure requires a later active curvature $q > p$ of the opposite sign. There are three possible selected locations. The first opposite point is $q \geq p+1 \geq r+2$. A single eligible zero between the two signs is selected at one index after the last active point, again at least $p+1$. On a longer zero plateau, the last point on the departing side is eligible only when it is not the first active point; hence it is at least $p+1$. Otherwise the arriving side is selected. The implementation's final spacing guard enforces the same lower bound in degenerate branches. Thus every nonterminal case has $t - r \geq 2$. □

Lemma 2 (Halving recurrence)**.** *If one level starts with $N \geq 1$ eligible intervals and produces N' chord intervals, then $N' \leq \lceil N/2 \rceil$.*

Proof. The produced intervals are consecutive and partition the N old intervals. By Lemma 1, every produced interval except possibly the last contains at least two old intervals. The last contains at least one. Hence $N \geq 2(N'-1)+1 = 2N'-1$, so $N' \leq (N+1)/2$ and integrality gives the claim. If the last interval also contains at least two, the stronger $N' \leq \lfloor N/2 \rfloor$ holds. □

Detailed completion of Theorem 2 and Corollary 1. Every new knot is selected from the current eligible list, so the mapped knot sets are nested. Let $N_0 = n-1$ and let N_j be the number of intervals after level j. Lemma 2 and the identity $\lceil \lceil a/2 \rceil /2 \rceil = \lceil a/4 \rceil$ give inductively $N_j \leq \lceil N_0/2^j \rceil$. For $n \geq 3$, choosing $j = \lceil \log_2(n-1) \rceil$ makes $N_j = 1$, so the endpoint chord has been recorded and the implementation stops. For $n = 2$, it records that chord in one level. This proves the stated $\max\{1, \lceil \log_2(n-1) \rceil\}$ bound. If a sparse implementation spends $O(N_j)$ work at level j, the geometric recurrence gives $O(n)$ total work. More explicitly, for integer $N_0 \geq 1$, $\lceil N_0/2^j \rceil \leq (N_0-1)/2^j + 1$, so $\sum_{j=1}^{J} N_j < N_0 - 1 + J \leq 2N_0$. Adding one endpoint per stored knot set gives $\sum_{j=1}^{J} |K_j| < 2N_0 + J$, the stated storage bound. This is the complexity of the implemented knot-only representation; materializing every length-n level remains $O(n \log n)$ in the worst case. □

Lemma 3 (Centred decision margin)**.** *On a uniform grid of spacing h, define γ and η as in Theorem 5. If $\|e\|_\infty < \min\{\gamma h/4, \eta h/8\}$, every curvature sign and every adjacent opposite-sign magnitude comparison made by the centred walk is unchanged.*

Proof. The divided-curvature perturbation stencil is $(1, -2, 1)/h$, so every coordinate changes by at most $4\|e\|_\infty/h$. The first strict inequality prevents zero crossing. For two curvature coordinates a, b, the reverse triangle inequality gives

$$|(|a+\delta a| - |b+\delta b|) - (|a| - |b|)| \leq |\delta a| + |\delta b| \leq 8\|e\|_\infty/h.$$

The second strict inequality therefore preserves every comparison outcome. Once signs are fixed there are no zero-curvature branches, and the potential comparison pairs are the adjacent entries across sign-run boundaries. These data determine the same knot walk. □

Detailed completion of Theorem 5. Lemma 3 fixes the knot set. At any sample between two fixed knots, the interpolation error is a convex combination of the two endpoint errors and is therefore at most $\|e\|_\infty$. This proves the baseline bound. The detail perturbation is $e - [T(y+e) - Ty]$, so the triangle inequality gives the factor two. If $\eta = 0$, no positive centred-rule radius follows: the explicit $(1, 1, -1)$ counterexample in the main text changes the selected side while preserving every sign. □

Lemma 4 (Essential class in the signed persistence metric)**.** *For a nonempty connected finite path, the ordinary zero-dimensional superlevel diagram has exactly one essential class. Under a finite-cost bottleneck matching between two functions on that path, the essential classes match each other; the remaining matching restricts to their finite diagrams.*

Proof. At the bottom of the filtration the entire path is present and connected, so exactly one component survives indefinitely. A diagram point with an infinite death coordinate has infinite distance from the diagonal and from every finite point. Therefore any finite-cost matching pairs the two essential points. Their cost is the absolute difference of their birth heights, which are the two function maxima. Removing that pair leaves precisely a matching of the finite parts. $\square$

Detailed completion of Theorem 8. The ith divided-curvature row is

$$\left(h_{i-1}^{-1}, -(h_{i-1}^{-1} + h_i^{-1}), h_i^{-1}\right),$$

whose absolute row sum is $2(h_{i-1}^{-1} + h_i^{-1})$. Taking the maximum gives the exact induced ℓ_∞ norm C_x. Coordinatewise positive part is 1-Lipschitz, so both signed vertex functions differ by at most $C_x\|y - z\|_\infty$. Their piecewise-linear extensions over a common edge differ by a convex combination of endpoint differences and obey the same bound. Apply persistence-diagram stability to their negatives, which converts superlevel filtrations to sublevel filtrations without changing the sup norm. Lemma 4 identifies the essential and finite costs used in d_{SC}. Taking the maximum over signs proves the metric bound. Finally, a finite bar (b, d) has diagonal cost $(b - d)/2$; if its lifetime is strictly greater than twice the metric bound, a bottleneck matching attaining that bound cannot send it to the diagonal. $\square$

Domain detail for Theorem 7. For $n \geq 3$, the divided-curvature matrix has rank $n - 2$: given any target curvature vector, choose an initial value and slope and integrate the target curvatures successively to construct a preimage. Thus D is surjective. Since a proper ϕ has nonempty domain, $\phi \circ D$ is proper; it is closed and convex by continuity and linearity of D. Adding $\|y-z\|_2^2/2$ makes the objective coercive and strongly convex, so its minimizer exists and is unique. The standard proximal theorem now applies to $g = \phi \circ D$. If $P = \mathrm{prox}_g$ is firmly nonexpansive, then so is $I - P$: writing $d = y - z$ and $p = Py - Pz$, firm nonexpansiveness gives $\|p\|^2 \leq \langle p, d\rangle$, which directly implies

$$\|(I - P)y - (I - P)z\|^2 \leq \langle (I - P)y - (I - P)z, y - z\rangle.$$

Finally, $D\ell = 0$ gives $g(w + \ell) = g(w)$; translating the optimization variable establishes affine equivariance. The $n = 2$ case has the zero curvature map and is immediate. $\square$

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